

ANX-PR/CL/001-01

LEARNING GUIDE

SUBJECT

595040537 - Introduction To Quantum Computing

DEGREE PROGRAMME

59ET - Doble Grado En Ing.Electronica De Comunicaciones Y En Ing.Telematica

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 2

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1. Description

1.1. Subject details

Name of the subject	595040537 - Introduction To Quantum Computing
No of credits	4.5 ECTS
Type	Optional/elective
Academic year of the programme	Third year
Semester of tuition	Semester 6
Tuition period	February-June
Tuition languages	English
Degree programme	59ET - Doble Grado en Ing.electronica de Comunicaciones y en Ing.telematica
Centre	59 - E.T.S. De Ingeniería Y Sist. De Telecom.
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Rafael Jose Hernandez Heredero (Subject coordinator)		rafael.hernandez.heredero@upm.es	Sin horario.
Rafael Delgado Lopez		rafael.delgado@upm.es	- -

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Prior knowledge recommended to take the subject

3.1. Recommended (passed) subjects

- Calculo I
- Algebra Lineal
- Calculo Ii
- Estadística Y Procesos Estocasticos

3.2. Other recommended learning outcomes

- This course cannot be followed and passed without a previous knowledge of Linear Algebra, Calculus I, Calculus II and some probability theory. They are not marked in this guide as compulsory prerequisites only because of administrative constraints.

4. Skills and learning outcomes *

4.1. Skills to be learned

CE B1 - Capacidad para la resolución de los problemas matemáticos que puedan plantearse en la ingeniería. Aptitud para aplicar los conocimientos sobre: álgebra lineal; geometría; geometría diferencial; cálculo diferencial e integral; ecuaciones diferenciales y en derivadas parciales; métodos numéricos; algorítmica numérica; estadística y optimización.

CE B2 - Conocimientos básicos sobre el uso y programación de los ordenadores, sistemas operativos, bases de datos y programas informáticos con aplicación en ingeniería.

CG 04 - Capacidad de abstracción, de análisis y de síntesis y de resolución de problemas.

CG 05 - Capacidad de trabajo en equipo y en entornos multidisciplinares.

CG 13 - Habilidades de aprendizaje con un alto grado de autonomía.

4.2. Learning outcomes

RA947 - To analyse and implement simple quantum circuits using simulation packages like Quirk and Qiskit.

RA945 - To compute the evolution of multi-qbit states, understanding its measurement and statistical interpretation.

RA948 - To understand quantum annealing techniques.

RA946 - To be acquainted with the basic types of quantum gates.

RA944 - To be familiar with applications of quantum computation to problems like the Satisfiability Problem.

RA949 - To know about some important quantum algorithms (quantum teleportation, Shor's algorithm, Grover's search).

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

5. Brief description of the subject and syllabus

5.1. Brief description of the subject

Brief description of the subject

This course is an introduction to quantum computing. We will explain the mathematical framework of quantum computation, beginning with quantum bits (qbits) and multi-qbits. The physical model of a qbit as a spin $\frac{1}{2}$ particle will be discussed, followed by an explanation of the unitary evolution and measurement procedure on a multi-qbit. Quantum gates and quantum circuits will be discussed, using simulators like qiskit and Quirk for designing and analysing simple cases. We will discuss several important quantum algorithms like quantum teleportation, Shor's factoring algorithm and Grover's search algorithm. We will end with an introduction to the already commercially available technology of quantum annealing, discussing its application to the problem of Satisfiability.

5.2. Syllabus

1. Introduction to quantum physics and quantum computing
 - 1.1. From the bit to the qbit
 - 1.2. Kroneker (or tensor) product
 - 1.3. The computational basis
2. Quantum states
 - 2.1. Quantum states, entanglement
 - 2.2. The spin 1/2 system. The Bloch sphere
 - 2.3. Hermitian operators and observables.
 - 2.4. Measurement
3. Quantum gates
 - 3.1. Introduction to Quantum Gates. Unitary operation over qubit states
 - 3.2. 1-qubit quantum gate: Pauli gate
 - 3.3. 2-qubits quantum gates: Hadamard and CNOT
 - 3.4. 3-qubits quantum gate: Toffoli gate
4. Quantum circuits
 - 4.1. Introduction to quantum gate notation and basic quantum circuits
 - 4.2. Usage of Quirk (Quantum Circuit Simulator)
 - 4.3. Construction of a Toffoli gate by means of CNOT gates
 - 4.4. Simple quantum circuits for increasing and decreasing
 - 4.5. A very basic introduction to the ?de-facto? standard (Python) qiskit framework
 - 4.6. Some quantum gate based algorithms: Quantum teleportation, Schorr?s algorithm and Grover?s search algorithm
5. Quantum annealing
 - 5.1. Introduction to the concept of quantum annealing
 - 5.2. Introduction to the Satisfiability Problem
 - 5.3. Choice of Hamiltonians and interpolation functions to solve a SAT problem via quantum annealing
 - 5.4. Simulation of simple quantum annealing in qiskit

6. Schedule

6.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	Chapter 1 Duration: 03:00 Lecture			
2	Chapter 2 Duration: 02:00 Lecture Chapter 2 Duration: 01:00 Problem-solving class			
3	Chapter 2 Duration: 02:00 Lecture Chapter 2 Duration: 01:00 Problem-solving class			Individual Work Individual presentation Progressive assessment Not Presential Duration: 00:00
4	Chapter 3 Duration: 03:00 Lecture			
5	Chapter 3 Duration: 02:00 Lecture Chapter 3 Duration: 01:00 Problem-solving class			
6	Chapter 4 Duration: 03:00 Lecture			
7		Chapter 4 Duration: 03:00 Laboratory assignments		
8	Chapter 4 Duration: 03:00 Lecture			
9		Chapter 4 Duration: 03:00 Laboratory assignments		
10		Chapter 4 Duration: 03:00 Laboratory assignments		Laboratory Evaluation Problem-solving test Progressive assessment Presential Duration: 03:00

11	Chapter 5 Duration: 03:00 Lecture			
12	Chapter 5 Duration: 03:00 Lecture			Group Work Group work Progressive assessment Not Presential Duration: 00:00
13		Chapter 5 Duration: 03:00 Laboratory assignments		
14		Chapter 5 Duration: 03:00 Laboratory assignments		
15				
16				
17				Global Exam Written test Progressive assessment Presential Duration: 03:00 Global Exam Written test Global examination Presential Duration: 03:00

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

7. Activities and assessment criteria

7.1. Assessment activities

7.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
3	Individual Work	Individual presentation	No Presential	00:00	10%	/ 10	CE B1 CG 04 CG 05 CG 13
10	Laboratory Evaluation	Problem-solving test	Face-to-face	03:00	25%	/ 10	CE B1 CE B2
12	Group Work	Group work	No Presential	00:00	15%	/ 10	CE B1 CG 04 CG 05 CG 13
17	Global Exam	Written test	Face-to-face	03:00	50%	3 / 10	CE B1 CG 04 CG 05 CG 13

7.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
17	Global Exam	Written test	Face-to-face	03:00	100%	5 / 10	CE B1 CE B2 CG 04 CG 05 CG 13

7.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
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Extraordinary Exam	Written test	Face-to-face	03:00	100%	5 / 10	CE B1 CE B2 CG 04 CG 05 CG 13
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7.2. Assessment criteria

Assessment criteria

There are two categories of graded items:

- Problem solving, group projects and lab activities: 50% of the total grade, as indicated above.
- Global Exam: the remaining 50%.

If the grading of the Global Exam results in a higher mark than the average between activities and exam, the final grade will be that of the exam. That is, the grade will be the result of applying the following formula:

Total grade=Max((activities+exam)/2, exam)

where both "activities" and "exam" are marked from 0 to 10 points. To pass the course, a total grade of 5 or more, and a minimum mark of 3 points in the global exam are required.

The extraordinary exam (in June/July) will allow to pass the course if a grade of 5 over 10 or more is obtained. The total grade will be that of the extraordinary exam.

8. Teaching resources

8.1. Teaching resources for the subject

Name	Type	Notes
N. D. Mermin. Quantum Computer Science, Cambridge University Press	Bibliography	Good reference for Chapters 1 to 4
M.A. Nielsen and I.L. Chuang. Quantum Computation and Quantum Information	Bibliography	Complete general reference (except for Chapter 5)
McMahon. Quantum Computing Explained	Bibliography	Simple introduction, accessible through UPM Ingenio
https://qiskit.org/learn/	Web resource	Nice textbook and tutorials by qiskit
https://qiskit.org/ecosystem/optimization/tutorials/index.html	Web resource	Introduction to quantum annealing

9. Other information

9.1. Other information about the subject

Sustainable Development Goals contribution.

The main SDG that the course can contribute to are, perhaps, the following.

SDG 4: Quality Education, by providing high level technical and theoretical knowledge promoting equality and sustainable development.

SDG 9: Industries, Innovation and Infrastructure by enhancing scientific research, and upgrading the technological capabilities of industrial sectors.

Prevalence of the 59ET Guide

Due to difficulties related to administrative and computing issues, there could be non-intended differences between the learning guides of subject "595040537 - Introduction To Quantum Computing". If this were the case, the "59ET - Doble Grado en Ing.electronica de Comunicaciones y en Ing.telematica" learning guide will prevail. The relevant exception is sections 3 and 4. For such sections the relevant guide in case of discrepancies would be "59ET - Doble Grado en Ing.electronica de Comunicaciones y en Ing.telematica".